

Camera-Based Driver Alertness Monitoring Using Object Detection

Shihab Uddin
ID: 011221017

Md. Ashmam Zamam Apon
ID: 011221157

Mhamudul Hasan Sadi
ID: 011221214

Md. Nazibullah
ID: 011221448

Tasnova Jahan Momo
ID: 0112230541

Mr. Shoib Ahmed Shourav
Supervisor

Dept. of Computer Science and Engineering
United International University
Dhaka, Bangladesh
suddin221017@bscse.uiu.ac.bd

Abstract—In this paper, seven classes (Drinking, Eating, Normal Focused, Normal Non-Focused, Sleeping, Using Phone, and Yawning/Drowsiness) of a camera-based driver behavior detector are evaluated. The experiments feature a split of YOLOv26n into day and night, as well as a comparison of four models taken from different architectures on a merged test set of 685 images. YOLOv26n achieved mAP@50/mAP@50-95 of 0.762/0.616 on the day test split and 0.512/0.455 on the night test split. On the common merged test set, YOLOv9 produced the highest mAP@50 (0.814) and mAP@50-95 (0.676), YOLOv10 the highest precision (0.800), and YOLOv11 the highest recall (0.784). Another training run of merged YOLOv26n achieved validation precision 0.965, recall 0.981, mAP@50 0.988 and mAP@50-95 0.923 at epoch-10 with an unknown number of samples, is not compared to the 685-image test results. There were differences in the scores from the day and night test split in particular, especially for Normal Focused recall and day Yawning/Drowsiness detection, but it doesn't isolate the effect of illumination. Times for model inference range from 1.6 to 6.0ms per image, not including the time required for loading the full application and alert. These image-level results are representative of the detector; it is yet to be determined if the continuous video warning performance is equal.

Index Terms—driver monitoring, driver drowsiness, object detection, YOLOv26n, day and night evaluation, computer vision

I. INTRODUCTION

Fatigue, drowsiness and distraction are important factors to consider as they can affect the driver's ability to react to events that are out of their control. Camera-based monitoring is a non-invasive approach to recognizing visible behaviors, without the need of wearable sensors. This work has been designed to classify seven different driver behavior classes from image in this project: Drinking, Eating, Normal Focused, Normal Non-Focused, Sleeping, Using Phone, Yawning/Drowsiness. The detector outputs class and bounding-box information, and the results measured in this paper are the image-level object detection metrics. These experiments don't measure event-level video performance or alert effectiveness, and the overall system design also features temporal decision logic and driver alerts.

A. Problem Statement

According to the statistics on the global road safety, many road traffic accidents are caused by the driver's carelessness, such as drowsy, tired and distracted driving. Past approaches to fatigue detection have mainly been based on physiological sensors, but they are expensive and inconvenient to use in common driving situations. In addition, a single measure, such as eye closure, is not always reliable, as it may result in false positives or not include more complicated distractive acts such as yawning or looking away from the road. Thus a system is desperately needed to track the level of alertness of the driver in real time which will ensure that potential collision is prevented and lives are saved without becoming too expensive and complex.

B. Contributions

The main contributions to be reported here are: (1) an object-detection formulation that covers seven classes of driver behaviors; (2) day and night aspects of YOLOv26n evaluations with per-class results; (3) a comparison of YOLOv9, YOLOv10, YOLOv11 and YOLOv12 using a common 685-image merged test set; (4) a separate merged YOLOv26n training and validation run with its configuration and learning results; and (5) an analysis of limitations at the class level and lighting aspects. The reported detection metrics do not validate temporal alerts and emergency communication, which are still considered to be system-design objectives.

II. RELATED WORK

Physiological sensing, computer vision and machine learning have developed to become a new tool for drowsiness detection. Machine-learning and vision-based fatigue recognition has gained in popularity in recent years, as summarized in recent reviews [1], [5]–[7]. Physiological and behavioral indicators of fatigue have also been studied using reaction time, alertness, eye closure, and blink duration [8]–[11]. Yawning has been used for visual drowsiness detection [2] and facial landmarks have been used for eye-blink detection [3] and for

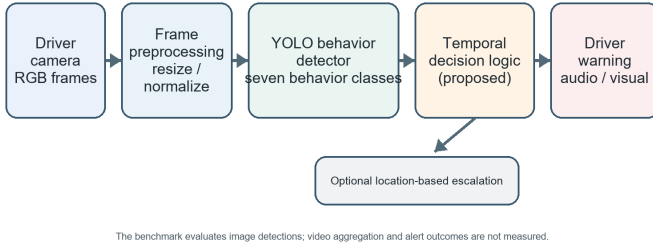


Fig. 1. Camera-based behavior detection and proposed alert flow.

eye-blink- and head-pose-recognition-based driver monitoring systems [4], [12], [13]. Camera-based systems have further examined facial appearance, gaze, and head pose under realistic driving conditions [14]–[17]. The value of complementary behavioral cues is demonstrated by these methods, however, depending on lighting, camera setup, accessories, and driving skill differences, performance may be impacted. This project proposes an alternate approach: an object detector which classifies 7 distinct states of the driver, including those of drowsiness and distraction. Temporal alerting is considered as a system design feature here, and the reported experiments evaluate image-level rather than a continuous driver-state decision. The detector implementation is related to established one-stage object-detection methods [18], [19] and deep residual feature extraction [20].

III. METHODOLOGY

The system is structured as a camera-to-alert pipeline: camera capture, frame preparation, object detection, behavior level decision logic and driver alert. In this paper, experiments were performed and assessed for the object-detection stage, with seven driver behaviors labeled. The overall architecture of the system is summarized in Fig. 1 and the context and data flow of the system is depicted in Fig. 2.

The camera sends the driver images to the detector which sends back the bounding boxes, class labels and confidence scores. The seven classes are not eye landmarks or mouth aspect ratios or head-pose angles but are observable behaviors. A downstream application could sum up detections over time to trigger an alarm and this temporal logic is not included in the image level benchmark and is a function of the system design.

A. Detection Pipeline

YOLO object detector is used on each of the input images. The YOLOv26n versions of the models run reported boxes and class scores as they are predicted. No separate face landmark extractor or EAR/MAR calculation or face pitch/yaw/roll calculation is used in the experiments.

B. Behavior Classes and Data

The detector detects the following: Drinking, Eating, Normal Focused, Normal Non-Focused, Sleeping, Using Phone, and Yawning/Drowsiness. The provided day yolo-v26n has

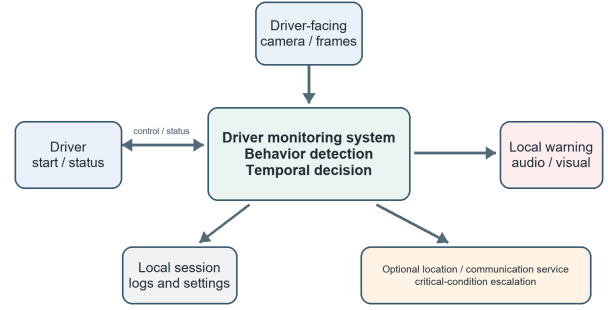


Fig. 2. System context with optional location-based escalation.

391 training split images and 380 instances, as well as 500 test images and 500 instances. There are 126 images and 126 instances reported in the night outputs and 185 images and 185 instances reported in the test outputs. The comparison of the merged YOLOv9–YOLOv12 models includes 685 test images and 685 instances.

C. Training Configuration

The merged run config contains a pretrained yolo model (yolov26n.pt), 10 epochs, 640-pixel image size, a batch size of 16, and validation during training. According to its model summary it has a total of 2,376,201 parameters, 122 layers and 5.2 GFLOPs. YOLOv11n has an output of 101 layers, 2,583,517 parameters and 6.3 GFLOPs. The day/night YOLOv26n result images were reported with the split names as the name of the file; their configuration files were not included with the result images.

D. Metrics

Precision is the percentage of correct predictions, recall is the percentage of labeled instances that are predicted. mAP@50 is the mean average precision at IoU (intersection over union) threshold of 0.50. The Precision at 50-95 (mAP) is the average of precision at all IoU thresholds in the range 0.50 to 0.95. Class support is the number of class instances labeled.

E. Runtime and Alert Scope

The provided logs contain details of preprocessing, inference and post-processing time for each image. These component timings represent model evaluation on the reported hardware only without any camera capture, temporal aggregation, U-I work and alert delivery. The overall design will include audio-visual warnings and/or an external communications option following an unsafe condition that persists but the present metrics will not provide evidence of the reliability or response time of those functions.

IV. SYSTEM DESIGN AND IMPLEMENTATION

The prototype suggested here is based on a forward-facing camera, image processing, YOLO behavior detector and an alert module at the end of the system. The following table (Table I) lists the planned hardware and software requirements.

TABLE I
SOFTWARE AND HARDWARE REQUIREMENTS FROM THE SYSTEM DESIGN

Component	Requirement
Operating system	Android 8.0 or higher
Programming	Python; Java or Kotlin
Computer vision	OpenCV; Ultralytics YOLO object detection
Machine learning	YOLOv26n trained and evaluated with Ultralytics/PyTorch
Storage	SQLite for session logs; Shared Preferences for settings
Camera	Front facing camera; minimum 720p; 15 to 30 FPS
Processing	Quad core processor or higher
Memory	Minimum 2 GB; 4 GB recommended
Output	Built in speaker; vibration optional
Location	GPS or device location service

The experiments reported here assess the performance of the image-based detection, and do not make statements about the performance of alerts from the mobile-device or end-to-end.

The design consists of acquisition of camera images, preparation of images, inference on the detector, aggregation of the behavior, and output of an alert. The detector module can be changed or retrained without any change of the application logic. The model outputs evaluated are class labels and bounding boxes, and temporal aggregation is necessary for the frame-level outputs to be a representation of a sustained driver state.

A. Functional Requirements

On startup of the monitoring, the application should start the camera and continuously check the driving session for the presence of the behavior detector. It should give warnings without the need for engagement by the vehicle while the vehicle is moving. The proposed interface is in Bengali as default and English as an optional language.

B. Non-Functional Requirements

Low latency, reliability, usability, affordability and maintainability are considered as non functional requirements. The results of the day vs. night test indicate that the accuracy will vary depending on the evaluation condition, and performance in a variety of scenes with different illumination, camera angles, motion blur and driver appearances needs to be validated. Not per-image inference time only gives end-to-end real-time behaviour.

V. EVALUATION FRAMEWORK AND RESULTS

Precision, recall, mAP@50 and mAP@50-95 were used to measure the evaluation of the detector. Also, YOLOv26n was evaluated on the provided day/night split sets while YOLOv9, YOLOv10, YOLOv11, and YOLOv12 were tested on a merged test set of 685 images. The validation results for every 10 epochs are presented in a separate merged YOLOv26n training log. The final values of that log for the epochs are not considered a matched test result as it was not provided the number of validation samples. The data in Tables II–VIII are rounded to three decimal places.

TABLE II
SUMMARY OF REPORTED DETECTION RESULTS

Experiment	Images/inst.	P	R	mAP50	mAP50-95
Day YOLOv26n train	391/380	0.972	0.973	0.985	0.946
Day YOLOv26n test	500/500	0.630	0.610	0.762	0.616
Night YOLOv26n train	126/126	0.883	0.940	0.975	0.861
Night YOLOv26n test	185/185	0.633	0.450	0.512	0.455
Merged YOLOv9 test	685/685	0.743	0.739	0.814	0.676
Merged YOLOv10 test	685/685	0.800	0.721	0.800	0.602
Merged YOLOv11 test	685/685	0.685	0.784	0.784	0.609
Merged YOLOv12 test	685/685	0.665	0.601	0.690	0.553
Merged YOLOv26n val e10	NR	0.965	0.981	0.988	0.923

TABLE III
REPORTED INFERENCE STAGE LATENCY IN MILLISECONDS PER IMAGE

Run	Pre	Infer	Post	Total
Day YOLOv26n train	0.2	1.6	0.4	2.2
Day YOLOv26n test	1.0	2.6	0.2	3.8
Night YOLOv26n train	0.1	1.6	0.2	1.9
Night YOLOv26n test	1.1	3.5	0.3	4.9
Merged YOLOv9	2.0	6.0	1.9	9.9
Merged YOLOv10	0.8	3.2	0.7	4.7
Merged YOLOv11	0.9	2.5	2.8	6.2
Merged YOLOv12	0.9	4.0	3.0	7.9

TABLE IV
YOLOv26N DAY TRAIN-SPLIT CLASS RESULTS

Class (instances)	P	R	mAP50	mAP50-95
Drinking (n=73)	0.981	0.904	0.987	0.949
Eating (n=23)	0.983	1.000	0.995	0.989
Normal Focused (n=40)	0.982	0.975	0.994	0.978
Normal Non-Focused (n=40)	0.987	1.000	0.995	0.993
Sleeping (n=50)	0.981	0.980	0.994	0.909
Using Phone (n=97)	1.000	0.910	0.950	0.954
Yawning/Drowsiness (n=57)	0.890	0.982	0.937	0.846

On the 685-image merged test set, YOLOv9 had the highest mAP@50 (0.814) and mAP@50-95 (0.676), YOLOv10 had the highest precision (0.800), and YOLOv11 had the highest recall (0.784). For YOLOv26n, test mAP@50 fell from 0.762 in the day split to 0.512 in the night split, while mAP@50-95 fell from 0.616 to 0.455. The high train-split scores and lower test scores do not mean that the performance is poor in unseen drivers or video, rather, they are a sign of generalization gap. The merged YOLOv26n log reached validation precision 0.96456, recall 0.98103, mAP@50 0.98807, and mAP@50-95 0.92346 at epoch 10. No reported sample count in the validation result and it is not compared with the matched test result.

Pre-processing, inference and post-processing are combined to make total. The supplied logs include loss time of 0.0 ms/image (not included in overall loss time). If the validation sample count is not reported in the merged log (YOLOv26n), it will be shown as "NR" in Table II.

The YOLOv26n results are given in tables IV–VII by class. The image and instance counts displayed are equal for these rows and the n value is the number of instances of the class.

In the combined test comparison, there is a variation in class

TABLE V
YOLOV26N DAY TEST-SPLIT CLASS RESULTS

Class (instances)	P	R	mAP50	mAP50-95
Drinking (n=100)	0.371	0.990	0.866	0.766
Eating (n=50)	0.975	0.783	0.981	0.830
Normal Focused (n=50)	1.000	0.000	0.663	0.471
Normal Non-Focused (n=50)	0.896	0.515	0.821	0.811
Sleeping (n=50)	0.763	1.000	0.988	0.689
Using Phone (n=100)	0.575	0.970	0.908	0.705
Yawning/Drowsiness (n=100)	0.062	0.010	0.109	0.024

TABLE VI
YOLOV26N NIGHT TRAIN-SPLIT CLASS RESULTS

Class (instances)	P	R	mAP50	mAP50-95
Drinking (n=15)	0.923	1.000	0.979	0.896
Eating (n=13)	0.699	1.000	0.924	0.719
Normal Focused (n=19)	0.807	1.000	0.990	0.911
Normal Non-Focused (n=17)	0.938	0.897	0.980	0.892
Sleeping (n=23)	0.898	1.000	0.950	0.864
Using Phone (n=19)	0.916	0.947	0.984	0.860
Yawning/Drowsiness (n=20)	1.000	0.737	0.976	0.883

TABLE VII
YOLOV26N NIGHT TEST-SPLIT CLASS RESULTS

Class (instances)	P	R	mAP50	mAP50-95
Drinking (n=20)	0.411	0.400	0.374	0.304
Eating (n=20)	0.565	0.350	0.368	0.320
Normal Focused (n=20)	1.000	0.000	0.179	0.158
Normal Non-Focused (n=25)	0.365	0.720	0.562	0.491
Sleeping (n=20)	0.446	0.950	0.907	0.850
Using Phone (n=40)	0.955	0.529	0.849	0.744
Yawning/Drowsiness (n=40)	0.691	0.200	0.346	0.306

TABLE VIII
YOLOV9-YOLOV12 CLASS RESULTS ON THE MERGED TEST SET

Class	Model	n	P	R	mAP50	mAP50-95
Drinking	YOLOv9	120	0.641	0.567	0.684	0.596
	YOLOv10	120	0.813	0.580	0.747	0.642
	YOLOv11	120	0.824	0.733	0.850	0.732
	YOLOv12	120	0.966	0.701	0.880	0.773
Eating	YOLOv9	70	0.863	0.901	0.926	0.801
	YOLOv10	70	0.843	0.900	0.924	0.761
	YOLOv11	70	0.950	0.857	0.892	0.769
	YOLOv12	70	1.000	0.784	0.928	0.752
Normal Focused	YOLOv9	70	0.798	0.714	0.838	0.750
	YOLOv10	70	0.961	0.711	0.744	0.614
	YOLOv11	70	0.911	0.731	0.885	0.591
	YOLOv12	70	0.291	0.025	0.398	0.273
Normal Non-Focused	YOLOv9	75	0.620	0.933	0.878	0.780
	YOLOv10	75	0.652	0.893	0.900	0.612
	YOLOv11	75	0.483	0.834	0.619	0.510
	YOLOv12	75	0.418	0.907	0.523	0.448
Sleeping	YOLOv9	70	0.820	0.871	0.925	0.677
	YOLOv10	70	0.899	0.762	0.920	0.639
	YOLOv11	70	0.311	0.782	0.794	0.579
	YOLOv12	70	0.704	0.857	0.916	0.666
Using Phone	YOLOv9	140	0.913	0.823	0.958	0.807
	YOLOv10	140	0.871	0.571	0.824	0.668
	YOLOv11	140	0.811	0.936	0.951	0.775
	YOLOv12	140	0.835	0.607	0.774	0.649
Yawning/Drowsiness	YOLOv9	140	0.545	0.359	0.490	0.318
	YOLOv10	140	0.580	0.629	0.538	0.276
	YOLOv11	140	0.505	0.612	0.494	0.310
	YOLOv12	140	0.441	0.321	0.409	0.310

TABLE IX
MERGED YOLOV26N VALIDATION METRICS BY EPOCH

Epoch	P	R	mAP50	mAP50-95
1	0.630	0.703	0.736	0.626
2	0.858	0.752	0.882	0.774
3	0.894	0.884	0.950	0.850
4	0.877	0.875	0.949	0.863
5	0.887	0.910	0.945	0.859
6	0.914	0.919	0.972	0.889
7	0.955	0.970	0.985	0.923
8	0.948	0.961	0.982	0.916
9	0.960	0.974	0.986	0.927
10	0.965	0.981	0.988	0.923

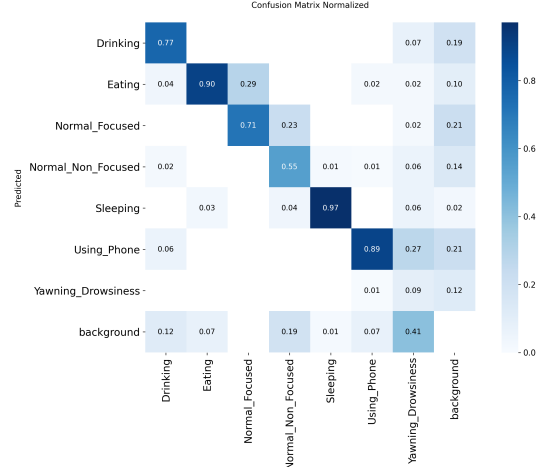


Fig. 3. Normalized confusion matrix for the merged YOLOv26n validation run.

performance by model. Yawning/Drowsiness is comparatively harder for all four models and Sleeping also has a low precision value (0.311) for YOLOv11 with a good recall value (0.782). Another drawback of YOLOv12 is its low Normal Focused recall (0.025). These class level results can be used in conjunction with aggregate mAP to choose a detector, for reporting precision and recall.

In the separate merged YOLOv26n run, the losses have reduced after 10 epochs of training: box loss is reduced from 0.597 to 0.296, the classification loss is reduced from 8.731 to 0.185 and the DFL loss is reduced from 0.02082 to 0.01010. Validation losses also declined: box loss from 0.648 to 0.332, classification loss from 4.095 to 0.198, and DFL loss from 0.02799 to 0.01322. Validation mAP@50 increased from 0.736 at epoch 1 to 0.988 at epoch 10. The validation metrics for each epoch are listed in Table IX - epoch 9 had the highest mAP@50-95 (0.927) with a slight edge over epoch 10 (0.923). The normalized confusion matrix in Fig. 3 summarizes class confusions for this validation run. It doesn't report the number of samples with the provided files.

VI. DISCUSSION

The results indicate that different detectors work best according to the way the evaluation is split (i.e., the target metric). In terms of mAP@50 and mAP@50-95, YOLOv9

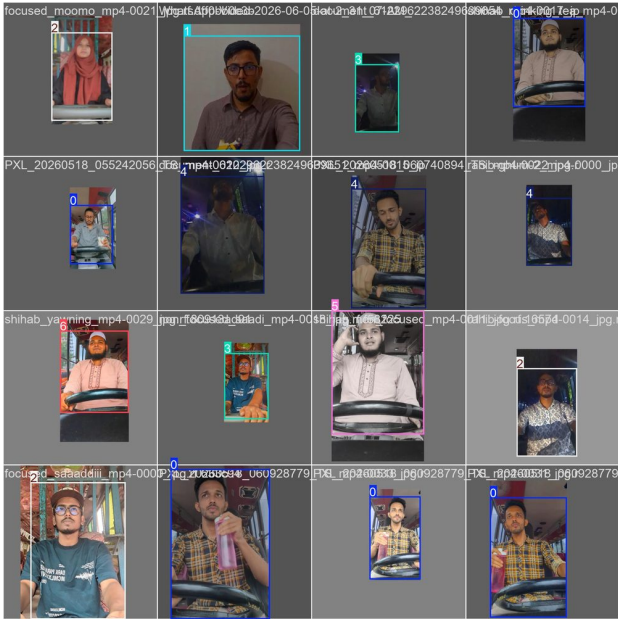


Fig. 4. Representative training batch used to illustrate the annotated driver-behavior examples supplied to the detector during training.

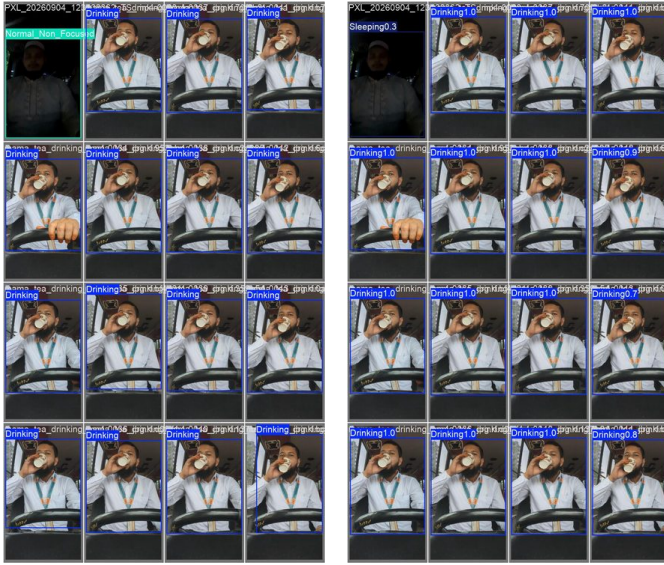


Fig. 5. Validation batch 0: ground-truth labels (left) and detector predictions (right).

performed best, while in precision, YOLO10 showed the best performance, and in recall, YOLOv11 was the best model, among the YOLOv9–YOLOv12, on the same merged test set of 685 images. YOLOv26n's day and night test splits showed that it achieved higher scores on the day split, with the night score being lower than the day score on the supplied training splits, and the night mAP being lower than the day mAP. This comparison is between the day and day split, and the night and night split, and isn't necessarily a difference in illumination as cause. The most serious flaws are class-specific: The recall for Normal focused is 0.0, the recall for yawning/drowsy at

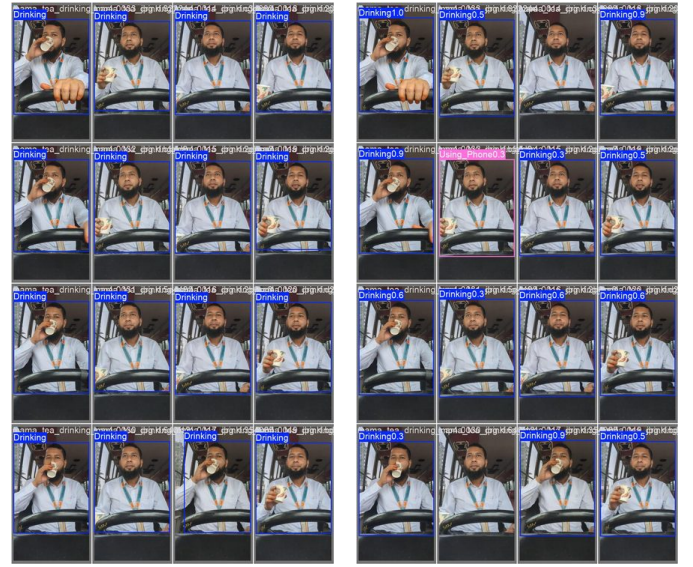


Fig. 6. Validation batch 1: ground-truth labels (left) and detector predictions (right).

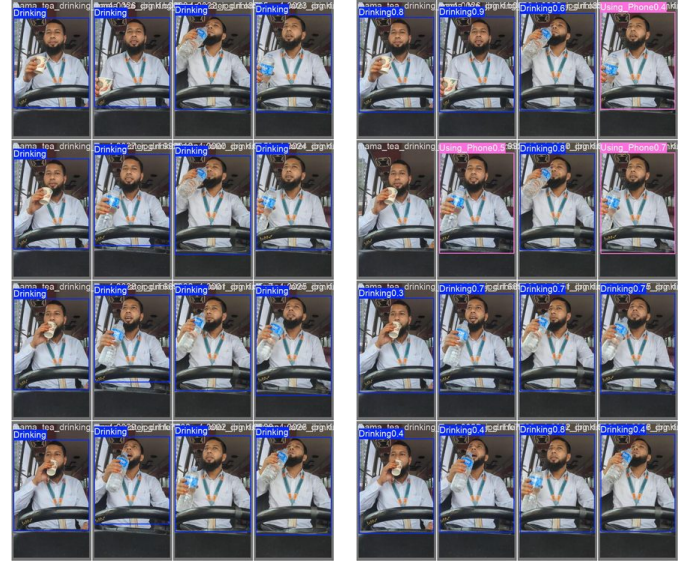


Fig. 7. Validation batch 2: ground-truth labels (left) and detector predictions (right).

daytime was 0.010, and at night the mAP at 50 was 0.346. Results indicate greater need for test examples that better represent the population, and for a better definition of lighting and driver variation classes, as well as across both lighting and driver variation classes. The separated merged yoloV26n epoch-10 validation result is encouraging but doesn't prove its superiority to the 685-image comparison, as it didn't provide the number of images or the test metrics used. Lastly, all reported measurements are in the image level detector result. State estimation (CV) and false-alert rate, response to alerts, and performance on the actual deployment platform are yet unmeasured.

VII. CONCLUSION

In this work, we analyze a camera based driver behavior detector with seven classes. On the common 685-image merged test set, YOLOv9 achieved the highest mAP@50 (0.814) and mAP@50-95 (0.676), YOLO10 the highest precision (0.800), and YOLOv11 the highest recall (0.784). Test mAP@50/mAP@50-95: 0.762/0.616 (day split), 0.512/0.455 (night split) and zero Normal Focused recall in both test mAP and test mAP 50-95, as well as poor day Yawning/Drowsiness test accuracy of 0.512. The separate merged YOLOv26n run found validation mAP@50 0.988 and mAP@50-95 0.923 at epoch 10 with an unknown amount of validation samples, which is not directly comparable to the matched test runs. The results provide information on image-level detector performance, but not end-to-end real-time warning performance, and reveal class and lighting limitations. Future study is needed to assess continuous video, false alert rate, response time and test the detector in different lighting, drivers, camera angles, and deployment hardware.

REFERENCES

- [1] R. Ali, Z. Abdelhalim, F. Abdallah, et al., "A systematic review on drivers drowsiness detection using machine learning: Awake behind the wheel," Research Square preprint, Dec. 2025.
- [2] S. Abtahi, B. Hariri, and S. Shirmohammadi, "Driver drowsiness monitoring based on yawning detection," in Proc. IEEE Int. Instrum. Meas. Technol. Conf., 2011, pp. 1–4.
- [3] J. Cech and T. Soukupova, "Real-time eye blink detection using facial landmarks," Central Machine Perception, Dept. of Cybernetics, Czech Technical University in Prague, 2016, pp. 1–8.
- [4] D. Kim, H. Park, T. Kim, W. Kim, and J. Paik, "Real-time driver monitoring system with facial landmark-based eye closure detection and head pose recognition," Scientific Reports, vol. 13, no. 1, 2023.
- [5] Y. Albadawi, M. Takruri, and M. Awad, "A review of recent developments in driver drowsiness detection systems," Sensors, vol. 22, no. 5, 2022.
- [6] G. Sikander and S. Anwar, "Driver fatigue detection systems: A review," IEEE Access, vol. 6, pp. 23664–23691, 2018.
- [7] J. S. Wijnands, C. W. Thompson, K. A. Nice, G. A. D. Kopp, and M. A. Z. A. T. Senserrick, "A review of driver drowsiness detection systems," IEEE Access, vol. 8, pp. 100282–100299, 2020.
- [8] D. F. Dinges and J. W. Powell, "Microcomputer analyses of performance on a portable, simple visual RT task during sustained operations," Behavior Research Methods, Instruments, and Computers, vol. 20, pp. 155–163, 1988.
- [9] T. Akerstedt, "Consensus statement: Fatigue and accidents in transport operations," Journal of Sleep Research, vol. 9, no. 4, pp. 395–398, 2000.
- [10] R. N. Khushaba, S. Kodagoda, S. Lal, and G. Dissanayake, "Driver drowsiness classification using fuzzy waveform-based feature extraction method," IEEE Transactions on Biomedical Engineering, vol. 58, no. 1, pp. 121–131, 2011.
- [11] S. K. L. Lal and A. Craig, "A critical review of the psychophysiology of driver fatigue," Biological Psychology, vol. 55, no. 3, pp. 173–194, 2001.
- [12] L. M. Bergasa, J. Nuevo, M. A. Sotelo, R. Barea, and M. E. Lopez, "Real-time system for monitoring driver vigilance," IEEE Transactions on Intelligent Transportation Systems, vol. 7, no. 1, pp. 63–77, 2006.
- [13] Q. Ji, Z. Zhu, and P. Lan, "Real-time nonintrusive monitoring and prediction of driver fatigue," IEEE Transactions on Vehicular Technology, vol. 53, no. 4, pp. 1052–1068, 2004.
- [14] M. J. Flores, J. M. Armingol, and A. de la Escalera, "Real-time drowsiness detection system for an intelligent vehicle," in Proc. IEEE Intelligent Vehicles Symposium, 2010, pp. 637–642.
- [15] T. Vincent, J. D. S. T. Silva, and A. L. C. Ottoni, "Drowsiness detection based on visual features and machine learning," in Proc. IEEE International Conference on Systems, Man, and Cybernetics, 2016, pp. 1–6.
- [16] J. Gwak, M. H. Shino, and A. H. Hwang, "Early detection of driver drowsiness using a camera-based system," in Proc. IEEE International Conference on Consumer Electronics, 2016, pp. 1–2.
- [17] A. Sahayadhas, K. Sundaraj, and M. Murugappan, "Detecting driver drowsiness based on sensors: A review," Sensors, vol. 12, no. 12, pp. 16937–16953, 2012.
- [18] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 779–788.
- [19] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," arXiv:2004.10934, 2020.
- [20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
- [21] Ultralytics, "YOLO documentation," 2023. [Online]. Available: <https://docs.ultralytics.com/>
- [22] J. Lee, S. Kim, and H. Kim, "Driver distraction detection using computer vision and gaze estimation," Sensors, vol. 21, no. 19, 2021.
- [23] Y. Dong, Z. Hu, K. Uchimura, and N. Murayama, "Driver inattention monitoring system for intelligent vehicles: A review," IEEE Transactions on Intelligent Transportation Systems, vol. 12, no. 2, pp. 596–614, 2011.
- [24] Q. Wang, M. Zhang, and J. Wang, "Driver attention monitoring using computer vision and deep learning," Neurocomputing, vol. 335, pp. 124–136, 2019.
- [25] M. Borghi et al., "Driver distraction detection using computer vision," Sensors, vol. 21, 2021.